

MAM2000W Module 2LA: Linear Algebra

Chapter 5

Some answers to the exercises

Section 5.1

1. This is similar to checking that the usual inner product on $\mathbb{R}^4$ is an inner product.
2. No. Axioms 1 and 4 hold, axioms 2 and 3 do not. (Give appropriate examples.)
3. (a) Yes. For any $p, q, r \in P_2$ and any scalar λ :
(1)

$$\begin{aligned}\langle p, q \rangle &= p(0)q(0) + p(1)q(1) + p(2)q(2) \\ &= q(0)p(0) + q(1)p(1) + q(2)p(2) \\ &= \langle q, p \rangle\end{aligned}$$

(2)

$$\begin{aligned}\langle p + q, r \rangle &= (p + q)(0)r(0) + (p + q)(1)r(1) + (p + q)(2)r(2) \\ &= p(0)r(0) + q(0)r(0) + p(1)r(1) + q(1)r(1) + p(2)r(2) + q(2)r(2) \\ &= p(0)r(0) + p(1)r(1) + p(2)r(2) + q(0)r(0) + q(1)r(1) + q(2)r(2) \\ &= \langle p, r \rangle + \langle q, r \rangle\end{aligned}$$

(3)

$$\begin{aligned}\langle \lambda p, q \rangle &= (\lambda p)(0)q(0) + (\lambda p)(1)q(1) + (\lambda p)(2)q(2) \\ &= \lambda(p(0)q(0) + p(1)q(1) + p(2)q(2)) \\ &= \lambda \langle p, q \rangle\end{aligned}$$

(4) $\langle p, p \rangle = p(0)p(0) + p(1)p(1) + p(2)p(2) = p^2(0) + p^2(1) + p^2(2)$. So $\langle p, p \rangle \geq 0$ for all $p \in P_2$. If p is the zero polynomial, then $\langle p, p \rangle = 0$. (Why?) Further, $\langle p, p \rangle = 0$ iff $p(0) = p(1) = p(2) = 0$. So then p is a quadratic that has three distinct roots (0,1 and 2) – that makes p the zero polynomial, as required. (This bit is crucial – make sure you understand it.)

(b) No. Axiom 4 fails.

4. Yes, it is an inner product on P_1 .
5. (a) 8 (b) -40 (c) $2\sqrt{53}$
6. $\|\lambda \mathbf{u}\|^2 = \langle \lambda \mathbf{u}, \lambda \mathbf{u} \rangle = \lambda \bar{\lambda} \langle \mathbf{u}, \mathbf{u} \rangle = |\lambda|^2 \|\mathbf{u}\|^2$, and the result follows on taking square roots.

7. The check is similar to the real case.
8. Use the fact that for any $\mathbf{w}$, $\|\mathbf{w}\|^2 = \langle \mathbf{w}, \mathbf{w} \rangle$ to write both sides of the identity in terms of inner products.

Section 5.2

1. (b) Draw a parallelogram with $\mathbf{u}$ and $\mathbf{v}$ as adjacent sides. The result in (a) says that the sides of this parallelogram have the same length if and only if its diagonals intersect at right angles.
2. No. $\langle (-1, 1, 0, 2), (1, -1, 3, 0) \rangle = -2$ not 0.
3. If $\mathbf{u} \in \text{span}(\{\mathbf{u}_1, \dots, \mathbf{u}_r\})$, then $\mathbf{u} = a_1\mathbf{u}_1 + \dots + a_r\mathbf{u}_r$. Now show that $\langle \mathbf{w}, \mathbf{u} \rangle = 0$.
4. (a) Each $\mathbf{u}_i \perp \mathbf{u}_j$ for $i \neq j$. An easy method is to use these vectors as columns of a matrix A . Then find $A^T A$. If $A^T A$ is diagonal, then each $\mathbf{u}_i \perp \mathbf{u}_j$ for $i \neq j$. (Why?)
 (b) $\mathbf{w}_1 \perp \mathbf{w}_2$, $\mathbf{w}_1 \perp \mathbf{w}_4$, $\mathbf{w}_2 \perp \mathbf{w}_4$.
5. Using the first inner product, we obtain:
 $\langle \mathbf{x}, \mathbf{y} \rangle = 3(\frac{1}{\sqrt{5}})(\frac{2}{\sqrt{30}}) + 2(-\frac{1}{\sqrt{5}})(\frac{3}{\sqrt{30}}) = 0$ and
 $\|\mathbf{x}\| = \sqrt{\langle \mathbf{x}, \mathbf{x} \rangle} = 3(\frac{1}{\sqrt{5}})(\frac{1}{\sqrt{5}}) + 2(-\frac{1}{\sqrt{5}})(-\frac{1}{\sqrt{5}}) = 1$.
 Similarly, $\|\mathbf{y}\| = 1$.
 Show yourself that $\langle \mathbf{x}, \mathbf{y} \rangle \neq 0$, $\|\mathbf{x}\| \neq 1$, $\|\mathbf{y}\| \neq 1$ if one uses the usual inner product on $\mathbb{R}^2$.

6. Applying the Gram-Schmidt procedure to the given vectors (in that order) gives:

$$\begin{aligned}
 \text{(a)} \quad & \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \frac{1}{\sqrt{6}} \begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix}, \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix}. \\
 \text{(b)} \quad & \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \frac{1}{\sqrt{6}} \begin{pmatrix} -2 & 1 \\ 1 & 0 \end{pmatrix}, \frac{1}{\sqrt{30}} \begin{pmatrix} 1 & 4 \\ -2 & -3 \end{pmatrix}. \\
 \text{(c)} \quad & \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{2}}x, \frac{\sqrt{6}}{2} \left(x^2 - \frac{2}{3} \right).
 \end{aligned}$$